



Ting Lin

📍 Shanghai, China ✉️ ting_lin@sjtu.edu.cn ☎️ +86 15821368966 in Ting-Lin
🌐 TtitiO 🎓 Ting Lin

Education

Shanghai Jiao Tong University

Sep 2021 - Aug 2025

UM-SJTU Joint Institute *BSE in Electrical and Computer Engineering*

- GPA: 3.74/4.0
- National Scholarship
- Excellent Undergraduate Graduate in Shanghai Jiao Tong University

Shanghai Jiao Tong University

Sep 2025 - June 2030 (expected)

Ph.D. student in Electrical Engineering

- Research Interest: Hardware-software co-design for efficient AI systems
- Advisor: Prof. Xinfei Guo

Publications

Cool-3D: An End-to-End Thermal-Aware Framework for Early-Phase Design Space Exploration of Microfluidic-Cooled 3DICs [🔗](#)

July 2025

Runxi Wang, Ziheng Wang, **Ting Lin**, Jacob M. Raby, Mircea R. Stan, Xinfei Guo
IEEE Journal on Emerging and Selected Topics in Circuits and Systems (JETCAS)

Shift-Left Techniques in Electronic Design Automation: A Survey [🔗](#)

September 2025

Xinyue Wu, Zixuan Li, Fan Hu, **Ting Lin**, Xiaotian Zhao, Runxi Wang, Xinfei Guo
arXiv preprint

Projects

3DIC Early-Phase Thermal Prediction Simulation — Advisor: Prof. Xinfei Guo

Aug 2024 - Mar 2025

- Engineered a robust data pipeline to seamlessly integrate the gem5 architectural simulator with the McPAT power model, automating the translation of performance statistics into accurate power inputs.
- Orchestrated the toolchain integration by resolving complex compilation, dependency, and compatibility issues between simulators, establishing a cohesive end-to-end simulation workflow for the Cool-3D framework.
- Developed and validated a meticulous data mapping scheme to accurately translate detailed microarchitectural events from gem5 (e.g., core activities, cache accesses) to McPAT's parameters, ensuring high-fidelity power estimation.

Approximate Logic Synthesis for Large-Scale Circuits — Advisor: Prof. Weikang Qian

Aug 2023 - Dec 2023

- Developed and tested two frameworks for 512-point Fast Fourier Transform (FFT) - Radix-2 and Mixed Radix-8x4x16.
- Implemented and validated a floating-point multiplier, enabling the integration of floating-point compatibility into FFT processes.
- Employed Approximate Logic Synthesis (ALS) algorithm for the synthesis of FFT circuit models, proving its effectiveness in managing and optimizing large-scale circuits.
- Integrated ALS algorithm into floating-point multiplication, resulting in a more compact circuit layout, reducing area, delay, and power consumption while maintaining computational accuracy, ensuring the functionality of floating-point FFT.

Technologies

Programming Languages: Proficient in Verilog/System Verilog, C++, Python, familiar with Triton, TileLang.

Project Development Technologies: Familiar with Linux and command line, proficient in using git for version control and team collaboration, and using Docker for production environment containerization; strong experience and ability in debugging and systematic problem solving.

EDA Tools: Familiar with EDA tools such as Synopsys Design Compiler & PrimeTime and AMD Vivado & Vitis.

Simulation Tools: Familiar with gem5 architectural simulator and McPAT power model.